Notebook 01 — Handover Document

Data Collection, Benchmarks & Crisis Analysis

What was done, what was produced, and what you need for continuation

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Table of Contents

1 Overview **2**

2 Raw Data Used **2**
- 2.1 Price Data — 11 US Assets (daily adjusted close prices) . . . 2
- 2.2 Volatility Signals . . . 2

3 What the Notebook Does — Part by Part **3**
- 3.1 Part A — Price Data Collection & Quality Control . . . 3
- 3.2 Part B — Volatility Signals (VIX & MOVE) . . . 3
- 3.3 Part C — Returns & Summary Statistics . . . 3
- 3.4 Part D — Benchmarks: Equal-Weight vs SPY . . . 3
- 3.5 Part E — Crisis Subsets . . . 4
- 3.6 Part F — Results & Visualisations . . . 4

4 What the Results Are Telling Us **4**

5 What You Should Use — Files & Paths **5**
- 5.1 Core Inputs — Load These at the Start of Your Notebook . . . 5
- 5.2 Crisis Subsets — Pre-sliced, Load Directly if Needed . . . 5
- 5.3 Summary Outputs (for Reference or Reporting) . . . 5
- 5.4 Key Constants to Reuse . . . 6

6 Recommended Workflow for Your Notebook **6**

7 Important Implementation Notes **6**

8 Summary — What Is Done vs What Remains **7**

Overview

This notebook (`01_data_collection_and_benchmarks.ipynb`) is the **foundation of the entire project pipeline**. It takes raw financial data as input, cleans and validates it, adds market volatility signals, computes returns and benchmarks, and analyses performance across five historical crisis periods. Everything produced here is saved to disk and is ready to be consumed directly by your notebook without re-running this one.

Raw Data Used

All raw files are located in `data/raw/`.

Price Data — 11 US Assets (daily adjusted close prices)

Ticker	File	Description
SPY	`spy_us_d.csv`	S&P 500 ETF — broad US equity market
QQQ	`qqq_us_d.csv`	Nasdaq-100 ETF — tech-heavy large cap
JPM	`jpm_us_d.csv`	JPMorgan Chase — US financials
XLI	`xli_us_d.csv`	Industrial Sector ETF
JNJ	`jnj_us_d.csv`	Johnson & Johnson — defensive healthcare
KO	`ko_us_d.csv`	Coca-Cola — defensive consumer staples
NEE	`nee_us_d.csv`	NextEra Energy — utilities
IEF	`ief_us_d.csv`	7–10Y US Treasury ETF — fixed income
GLD	`gld_us_d.csv`	Gold ETF — safe haven commodity
XLE	`xle_us_d.csv`	Energy Sector ETF
UPS	`ups_us_d.csv`	UPS — industrials/logistics

The asset universe was deliberately chosen to span **multiple asset classes and sectors** — equities (growth, defensive, financial, industrial), fixed income, commodities, and utilities — in order to study diversification effects under different market conditions.

Volatility Signals

File	Description
`VIX_History.csv`	CBOE VIX index — equity implied volatility, the standard equity "fear gauge"
`MOVE_History.csv`	ICE BofA MOVE index — bond implied volatility, the bond market equivalent of VIX

What the Notebook Does — Part by Part

Part A — Price Data Collection & Quality Control

Loads all 11 price CSVs, merges them into a single panel, clips to the common date range **2005-02-25 → 2026-04-17**, and runs three data quality checks:

- **Missing values** per asset — identifies dates with incomplete observations
- **Extreme daily returns (>30%)** — catches unadjusted corporate actions or data errors
- **Price jump anomalies (<0.6× or >1.8× previous close)** — catches unadjusted stock splits

Result: The data is clean. Only 6 NaNs were found on a single date (2011-02-17, Presidents' Day), handled by `dropna()`. Zero extreme return events. Zero unadjusted splits. Two calendar gaps >4 days, both corresponding to documented NYSE closures (New Year's 2007, Hurricane Sandy 2012). The final clean panel has **5,318 trading days × 11 assets**.

Part B — Volatility Signals (VIX & MOVE)

Loads VIX and MOVE, aligns them to the price panel's trading calendar using forward-fill, and merges everything into a single master panel. Both signals are saved and available as regime indicators — historically, VIX above 30 marks equity stress periods; elevated MOVE marks bond market dislocations.

Part C — Returns & Summary Statistics

Computes **daily log returns** for all 11 assets and derives full-period annualised statistics: mean return, volatility, Sharpe ratio, skewness, excess kurtosis, and max drawdown per asset. Log returns are used (not simple returns) because they are time-additive and more suitable for statistical modelling.

Notable findings: GLD shows positive skewness and low correlation with equities — consistent safe-haven behaviour. IEF has the lowest annualised volatility in the universe. XLI and XLE show the largest full-period drawdowns. Most equity assets exhibit negative skewness and excess kurtosis, confirming fat-tailed, crash-prone return distributions.

Part D — Benchmarks: Equal-Weight vs SPY

Constructs two passive buy-and-hold benchmarks starting from $1 in 2005:

- **Equal-Weight (EW):** Equal dollar allocation across all 11 assets at inception, no rebalancing
- **SPY Buy-and-Hold:** Full allocation to SPY only

Both benchmarks include a full-period wealth index, rolling 252-day Sharpe ratio, and drawdown series.

Full-period results (2005–2026):

	Ann. Mean	Ann. Vol	Sharpe	Total Return	Max Drawdown
EW Passive	10.89%	14.62%	0.745	+693%	−37.74%
SPY Buy-and-Hold	11.42%	19.06%	0.599	+658%	−56.47%

The EW portfolio achieves similar total return to SPY with significantly lower volatility and drawdown, resulting in a higher Sharpe ratio. This confirms that cross-asset diversification adds meaningful risk-adjusted value even without active management.

Part E — Crisis Subsets

Slices all data panels across five historically significant stress periods:

Crisis	Period	Days	Stress Type
GFC 2008	Oct 2007 – Mar 2009	378	Credit & liquidity collapse
EU Debt 2011	Apr 2010 – Jun 2012	567	Sovereign debt contagion
COVID 2020	Feb 2020 – Jun 2020	104	Exogenous shock & rapid recovery
Rate Hike 2022	Jan 2022 – Dec 2022	251	Monetary tightening cycle
Trump Tariffs 2025	Jan 2025 – Apr 2026	323	Geopolitical & trade uncertainty

For each crisis, four data panels are saved: prices, signals, returns, and benchmark series — all pre-sliced and immediately ready to load.

Part F — Results & Visualisations

Produces the final comparative analysis across all crisis windows.

EW vs SPY — Crisis Performance Summary:

Crisis	EW Total%	SPY Total%	EW vs SPY	EW Max DD	SPY Max DD
GFC 2008	−26.67%	−48.46%	**+21.79%**	−37.74%	−56.47%
EU Debt 2011	+26.49%	+19.13%	**+7.36%**	−10.61%	−18.62%
COVID 2020	−6.07%	−3.87%	**−2.20%**	−30.00%	−33.72%
Rate Hike 2022	−10.28%	−18.64%	**+8.36%**	−20.00%	−24.50%
Trump Tariffs 2025	+31.90%	+21.83%	**+10.07%**	−14.39%	−18.76%

Key finding: EW outperforms SPY in 4 out of 5 crisis periods with consistently lower drawdowns across all five. The only exception is COVID-2020, where SPY recovered faster due to its tech concentration (QQQ +13.5%). The asset heatmap confirms that GLD and IEF are the clearest safe-haven assets, while XLI and XLE are the most cyclically exposed.

What the Results Are Telling Us

The central message is that the Equal-Weight passive benchmark is **more resilient than SPY in most stress regimes**, and that the asset universe contains **meaningful cross-asset variation** across different types of crisis. This is not a coincidence of asset selection — it reflects the deliberate inclusion of defensive, fixed income, and commodity assets alongside equities.

From an asset-level perspective, the crisis heatmap shows that assets do not all move together. Defensive assets such as IEF and GLD act as stabilisers in several regimes, while sector behaviour shifts significantly depending on the type of shock. This variation is exactly what makes the universe useful for stress-based allocation logic — if all assets behaved the same way, no diversification benefit would be possible.

The practical implication for your work: the input data contains **real signal variation across crisis regimes**, which means your stress score will have something meaningful to detect and differentiate.

What You Should Use — Files & Paths

Core Inputs — Load These at the Start of Your Notebook

```
import pandas as pd

# Clean daily prices -- 5,318 rows x 11 assets
prices = pd.read_csv("data/processed/prices_clean.csv",
                     parse_dates=["Date"], index_col="Date")

# Aligned VIX + MOVE signals -- same date index as prices
signals = pd.read_csv("data/processed/signals.csv",
                      parse_dates=["Date"], index_col="Date")

# Master panel -- prices + signals merged on the same timeline
master = pd.read_csv("data/processed/master_prices_with_signals.csv",
                     parse_dates=["Date"], index_col="Date")

# Daily log returns -- 5,317 rows x 11 assets
log_returns = pd.read_csv("data/processed/log_returns.csv",
                          parse_dates=["Date"], index_col="Date")

# Benchmark panel -- EW & SPY wealth, returns, rolling Sharpe, drawdown
benchmarks = pd.read_csv("data/processed/benchmarks.csv",
                         parse_dates=["Date"], index_col="Date")
```

Crisis Subsets — Pre-sliced, Load Directly if Needed

```
# Replace {CRISIS} with one of:
# GFC_2008 | EU_Debt_2011 | COVID_2020 | Rate_Hike_2022 | Trump_Tariffs_2025

prices_crisis  = pd.read_csv("data/processed/crisis/08_1_{CRISIS}_prices.csv",
                             parse_dates=["Date"], index_col="Date")
signals_crisis = pd.read_csv("data/processed/crisis/08_2_{CRISIS}_signals.csv",
                             parse_dates=["Date"], index_col="Date")
returns_crisis = pd.read_csv("data/processed/crisis/08_3_{CRISIS}_returns.csv",
                             parse_dates=["Date"], index_col="Date")
stats_crisis   = pd.read_csv("data/processed/crisis/08_4_{CRISIS}_asset_stats.csv",
                             index_col=0)
```

Summary Outputs (for Reference or Reporting)

Description	Path
Full-period benchmark summary	`outputs/01_data_collection/04_benchmark_summary.csv`
Asset return stats (full period)	`outputs/01_data_collection/03_1_returns_summary_stats.csv`
EW vs SPY crisis report	`outputs/01_data_collection/14_final_crisis_report.csv`
Per-crisis bench summary (×5)	`outputs/01_data_collection/08_5_{CRISIS}_bench_summary.csv`

Key Constants to Reuse

```
HELD_ASSETS = ['SPY', 'QQQ', 'JPM', 'XLI', 'JNJ',
               'KO', 'NEE', 'IEF', 'GLD', 'XLE', 'UPS']
TRADING_DAYS = 252

CRISIS_WINDOWS = {
    "GFC_2008":           ("2007-10-01", "2009-03-31"),
    "EU_Debt_2011":       ("2010-04-01", "2012-06-30"),
    "COVID_2020":         ("2020-02-01", "2020-06-30"),
    "Rate_Hike_2022":     ("2022-01-01", "2022-12-31"),
    "Trump_Tariffs_2025": ("2025-01-01", "2026-04-17"),
}
```

Recommended Workflow for Your Notebook

Your role is to build the **stress detection and uncertainty/risk module**. Based on what is available from this notebook, the recommended workflow is:

1. **Load the master panel** (`master_prices_with_signals.csv`) — it already contains prices and aligned VIX/MOVE on the same index
2. **Compute price-derived stress indicators** from the asset price panel:
   - Realized volatility (rolling standard deviation of returns)
   - Downside returns (negative return days only)
   - Rolling drawdown / short-term drawdown from wealth
3. **Use VIX and MOVE directly** from the `signals` columns — no additional alignment needed
4. **Normalise all indicators** (e.g. z-scores over a rolling window)
5. **Combine into a single daily stress score**
6. **Validate your score** against the pre-sliced crisis subsets — your stress score should spike during GFC 2008, COVID 2020, and Rate Hike 2022 windows

Important Implementation Notes

- `prices_clean.csv` is the **single source of truth** for all price-based calculations. Always load from this file rather than re-processing the raw CSVs.
- Use `log_returns` for statistical modelling and `prices.pct_change()` (simple returns) for portfolio wealth construction — this is the convention used throughout this notebook.
- The `signals` DataFrame is already perfectly aligned to the price calendar — you can merge on the index without any date handling.
- The crisis subsets are fully self-contained — you do not need to re-define the windows or re-slice the data. Load them directly and use them for validation.
- The `benchmarks` DataFrame is your performance baseline — any active or signal-driven strategy you build should be compared against `ew_wealth` and `spy_wealth` from that file.

Summary — What Is Done vs What Remains

Status	Item
Done	Final held-asset universe (11 assets)
Done	Clean aligned price panel (5,318 days)
Done	Aligned VIX & MOVE signals
Done	Daily log returns & descriptive statistics
Done	Passive benchmark construction (EW & SPY)
Done	Crisis windows definition & data slicing
Done	Benchmark and crisis comparison results
Your task	Stress indicator engineering (realised vol, downside, drawdown)
Your task	Signal normalisation & daily stress score
Your task	Uncertainty/risk module construction
Your task	Score validation against crisis windows